

Median Filter

1. Sliding Window Animation

Consider a **5×5** image:

10 20 30 40 50

15 25 35 45 55

12 22 32 42 52

18 28 38 48 58

14 24 34 44 54

We apply a **3×3** median filter.

Window Position 1 (center pixel = 25)

Window over top-left:

10 20 30

15 25 35 ← center = 25

12 22 32

Sorted:[10,12,15,20,22,25,30,32,35]

Median = **22**

So output pixel at that position = **22**

Window moves right → Position 2 (center=35)

20 30 40

25 35 45

22 32 42

Sorted:[20,22,25,30,32,35,40,42,45]

Median = **32**

Output = **32**

The window continues sliding across the entire row... Then it moves to:

- Next row
- Next row
- Until all pixels are processed

This is called a **convolution sliding operation**.

Filtered 3×3 valid region output:

22 32 42

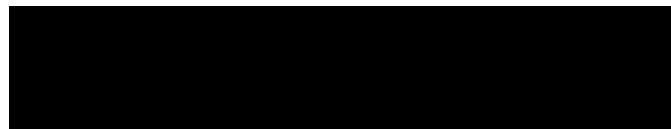
25 35 45

24 34 44

Every output pixel is the **median** of its own 3×3 neighborhood.

Summary:

- Median filter uses a sliding 3×3 window
- For each window, it finds the **median of 9 values**
- Only the **center pixel of that window** is replaced
- The window shifts one pixel at a time (ie The **window slides right**) ✓ After finishing the row, it **slides down**
 - ✓ Then slides right again
 - ✓ Repeat until the entire image is processed
- Over the whole image, **every pixel is updated once**
- Result: noise removed, edges preserved



1. Gaussian Kernel (Example 3×3, $\sigma \approx 1$)

A typical normalized 3×3 Gaussian kernel is:

$$\begin{bmatrix} \frac{1}{16} & \frac{2}{16} & \frac{1}{16} \\ \frac{2}{16} & \frac{4}{16} & \frac{2}{16} \\ \frac{1}{16} & \frac{2}{16} & \frac{1}{16} \end{bmatrix}$$

Gaussian = weighted average

Not like median (which only picks the middle value).

For each sliding 3×3 window:

Final pixel = $\text{Sum}(\text{pixel} \times \text{weight}) / \text{Sum}(\text{weights})$

5×5 image

10 20 30 40 50

15 25 35 45 55
12 22 32 42 52
18 28 38 48 58
14 24 34 44 54

We will compute Gaussian filtered values for the valid 3×3 centers

SLIDING WINDOW PROCESS

WINDOW 1

10 20 30
15 25 35
12 22 32

Multiply **Kernel** element-wise:

$$10 \times 1 = 10$$

$$20 \times 2 = 40$$

$$30 \times 1 = 30$$

$$15 \times 2 = 30$$

$$25 \times 4 = 100$$

$$35 \times 2 = 70$$

$$12 \times 1 = 12$$

$$22 \times 2 = 44$$

$$32 \times 1 = 32$$

$$\text{Sum all: } 10 + 40 + 30 + 30 + 100 + 70 + 12 + 44 + 32 = 368$$

$$\text{Normalize: } 368 / 16 = 23$$

Continue this process.....

FINAL GAUSSIAN FILTER OUTPUT:

23 33 43

24 34 44

26 36 46

Comparison With Median Filter Result

22 32 42

25 35 45

24 34 44

23 33 43

24 34 44

	Filter	How it works	Output	Nature
Median		Replaces with median	Sharp edges preserved	
Gaussian		Weighted average	Smooth, soft edges	